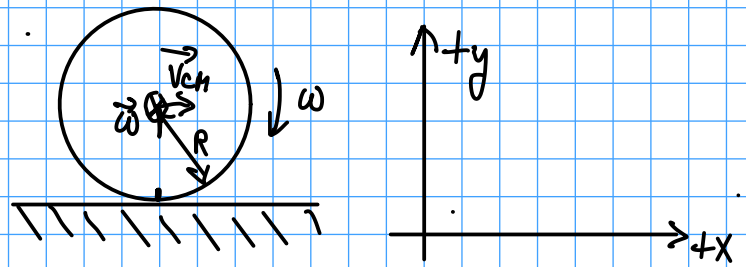


# SUMMARY ON ELEMENTARY ROTATIONS:

$$L_z = I \omega_z \quad (1) \quad \vec{N} = \vec{r} \times \vec{F} \quad (2) \quad T = \frac{I}{2} \omega_z^2 = \frac{\omega_z L_z}{2} \quad (3)$$

$$\frac{d\vec{L}}{dt} = \vec{N}^{(e)} \quad (4) \quad \begin{cases} \text{INERTIAL FRAME} \\ \text{CM FRAME (ORIGIN AT CM, AXES PARALLEL TO INERTIAL AXES)} \end{cases}$$

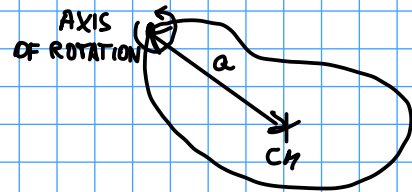
Rolling without slipping condition:  $v_{CM} + \omega R = 0$   $(5)$   
 $(\omega < 0 \text{ for clockwise rotation})$   $(6)$



Parallel Axis Theorem:  $I = \bar{I} + Ma^2$   $(7)$

$(8) \bar{I}$  = Moment of inertia around CM

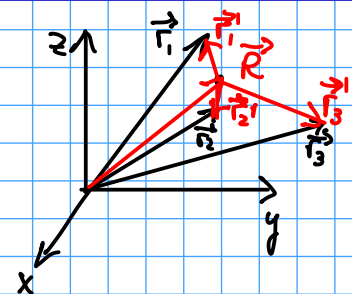
$(9) a$  = Distance between axis of rotation and CM



Separating CM Motion:  $M = \sum m_j$   $(12)$

$$\vec{L} = \vec{R} \times M \dot{\vec{R}} + \sum_j \vec{r}_j' \times m_j \dot{\vec{r}}_j' \quad (10) \quad \vec{R} = \frac{\sum m_j \vec{r}_j}{M} \quad (13)$$

$$T = \frac{M}{2} |\dot{\vec{R}}|^2 + \frac{1}{2} \sum_j m_j |\dot{\vec{r}}_j'|^2 \quad (11) \quad \vec{r}_j' = \vec{r}_j - \vec{R} \quad (14) \quad \text{POSITION RELATIVE TO CM}$$



For rigid body, considering rotation with angular velocity  $\omega_z$  around CM:

$$L_z = (\vec{R} \times M \dot{\vec{R}})_z + I \omega_z \quad (16) \quad T = \frac{M}{2} |\dot{\vec{R}}|^2 + \frac{I}{2} \omega_z^2 \quad (17)$$

## PROOFS:

CHANGING TO CM SYSTEM:

$$\vec{L} = \sum_{i=1}^N \vec{r}_i \times m_i \dot{\vec{r}}_i = \sum_{i=1}^N (\vec{R} + \vec{r}_i') \times m_i (\dot{\vec{R}} + \dot{\vec{r}}_i') = \sum_i \vec{R} \times m_i \dot{\vec{R}} + \sum_i \vec{r}_i' \times m_i \dot{\vec{r}}_i' + \text{CROSS-TERMS} \quad (18)$$

$$\vec{r}_i = \vec{R} + \vec{r}_i' \quad \dot{\vec{r}}_i = \dot{\vec{R}} + \dot{\vec{r}}_i' \quad (19)$$

$$\sum_i \vec{r}_i' \times m_i \dot{\vec{R}} = \sum_i m_i (\vec{r}_i' - \vec{R}) \times \dot{\vec{R}} = \left[ \sum_i m_i \vec{r}_i' - \left( \sum_i m_i \right) \vec{R} \right] \times \dot{\vec{R}} = [M\vec{R} - M\vec{R}] \times \dot{\vec{R}} = \vec{0} \quad (20)$$

$$\sum_i \vec{R} \times m_i \dot{\vec{r}}_i' = \vec{R} \times \sum_i m_i (\dot{\vec{r}}_i' - \dot{\vec{R}}) = \vec{R} \times \left[ \sum_i m_i \dot{\vec{r}}_i' - \left( \sum_i m_i \right) \dot{\vec{R}} \right] = \vec{R} \times [M\dot{\vec{R}} - M\dot{\vec{R}}] = \vec{0} \quad (21)$$

THEN:  $\vec{L} = \vec{R} \times M \dot{\vec{R}} + \sum_i \vec{r}_i' \times m_i \dot{\vec{r}}_i' \quad (22)$

$\Rightarrow \frac{d\vec{L}}{dt} = \vec{R} \times M \ddot{\vec{R}} + \sum_i \vec{r}_i' \times m_i \ddot{\vec{r}}_i' \quad (23) \quad \text{TIME DERIVATIVE}$

BUT ALSO:  $\frac{d\vec{L}}{dt} = \vec{N}^{(e)} = \sum_i \vec{r}_i \times \vec{F}_i^{(e)} \quad (24)$

$$\frac{d\vec{L}}{dt} = \vec{R} \times \sum_i \vec{F}_i^{(e)} + \sum_i \vec{r}_i' \times \vec{F}_i^{(e)} = \vec{R} \times M \ddot{\vec{R}} + \sum_{i=1}^N \vec{r}_i' \times \vec{F}_i^{(e)} \quad (26)$$

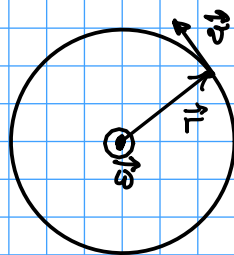
$(25) M\ddot{\vec{R}} = \sum_i \vec{F}_i^{(e)}$  Newton's 2nd Law for particle system

$$\frac{d\vec{L}'}{dt} = \sum_{i=1}^N \vec{r}_i' \times \vec{F}_i^{(e)} \quad (27)$$

Eq (27) is the special case of Eq (4) in the CM Frame.

# Rotations, Non-inertial systems, and Rigid body motion

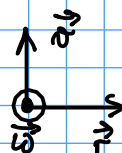
CIRCULAR MOTION  
IN A PLANE



$$\vec{v} = \vec{\omega} \times \vec{r}$$

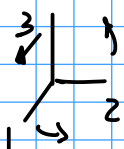
$$\vec{v} \perp \vec{r}$$

$$\vec{r} \cdot \vec{r} = r^2$$



$$\hat{e}_i \cdot \hat{e}_j = \delta_{ij} = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$$

ROTATING COORDINATE SYSTEM



$$0 = \frac{d\delta_{ij}}{dt} = \frac{d\hat{e}_i \cdot \hat{e}_j}{dt}$$

$$0 = \hat{e}_i \cdot \frac{d\hat{e}_j}{dt} \quad i=j$$

$$0 = \hat{e}_i \cdot \frac{d\hat{e}_j}{dt} + \hat{e}_j \cdot \frac{d\hat{e}_i}{dt} \quad i \neq j$$

$(\hat{e}_j)_{j=1,2,3}$  IS A BASIS  $\Rightarrow$  WE CAN WRITE  $\frac{d\hat{e}_i}{dt}$  AS A LINEAR COMBINATION

$$\frac{d\hat{e}_i}{dt} = \sum_{j=1}^3 \Omega_{ij} \hat{e}_j \quad \Omega_{ij} = \hat{e}_j \cdot \frac{d\hat{e}_i}{dt}$$

(USE THAT  $\hat{e}_j \cdot \sum_{i=1}^3 \Omega_{ij} \hat{e}_i = \sum_{i=1}^3 \Omega_{ij} \hat{e}_j \cdot \hat{e}_i = \Omega_{ij}$ )

$$\hat{x} \times \hat{y} = \hat{z}$$

$$\hat{y} \times \hat{z} = \hat{x}$$

$$\hat{z} \times \hat{x} = \hat{y}$$

$$\hat{e}_1 \equiv \hat{x} \quad \hat{e}_2 \equiv \hat{y} \quad \hat{e}_3 \equiv \hat{z}$$

$$\hat{e}_1 \times \hat{e}_2 = \hat{e}_3 \quad (\text{AND CYCLICALLY})$$

$$\hat{e}_2 \times \hat{e}_3 = \hat{e}_1$$

$$\hat{e}_3 \times \hat{e}_1 = \hat{e}_2$$

Levi-Civita symbol

$$\epsilon_{ijk} = \begin{cases} +1 & ijk \text{ cyc } 123 \\ -1 & ijk \text{ cyc } 213 \\ 0 & \text{if any repeated index} \end{cases}$$

$$\epsilon_{123} = \epsilon_{231} = \epsilon_{312} = +1$$

$$\epsilon_{213} = \epsilon_{132} = \epsilon_{321} = -1$$

$$\epsilon_{jji} = \epsilon_{ijj} = \epsilon_{jij} = 0$$

$$\hat{e}_i \times \hat{e}_j = \sum_l \epsilon_{ijl} \hat{e}_l$$

$$\langle \psi_i | \left( \sum_j c_j |\psi_j\rangle \right) = \sum_j c_j \langle \psi_i | \psi_j \rangle = c_i$$

$$i \neq j \quad 0 = \Omega_{ij} + \Omega_{ji} \Rightarrow \Omega_{ji} = -\Omega_{ij}$$

$$\Omega_{ii} = \hat{e}_i \cdot \frac{d\hat{e}_i}{dt} = 0$$

$$\begin{pmatrix} 0 & \Omega_{12} & -\Omega_{31} \\ -\Omega_{12} & 0 & \Omega_{23} \\ \Omega_{31} & -\Omega_{23} & 0 \end{pmatrix}$$

$$\omega_1 \equiv \Omega_{23} \quad \omega_2 \equiv \Omega_{31} \quad \omega_3 \equiv \Omega_{12}$$

$$\frac{d\hat{e}_1}{dt} = \Omega_{12} \hat{e}_2 + \Omega_{13} \hat{e}_3 = \Omega_{12} \hat{e}_2 - \Omega_{31} \hat{e}_3 = \omega_3 \hat{e}_2 - \omega_2 \hat{e}_3 = \vec{\omega} \times \hat{e}_1 \quad \checkmark$$

$$\vec{\omega} \times \hat{e}_1 = (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2 + \omega_3 \hat{e}_3) \times \hat{e}_1 = -\omega_2 \hat{e}_3 + \omega_3 \hat{e}_2$$

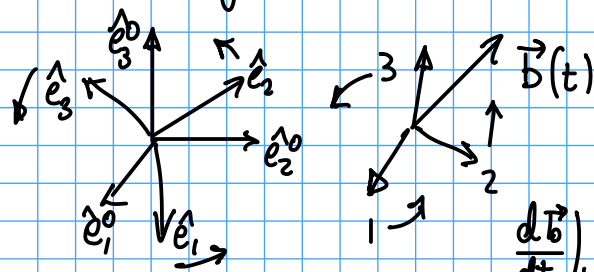
$$\frac{d\hat{e}_2}{dt} = \Omega_{21} \hat{e}_1 + \Omega_{23} \hat{e}_3 = -\Omega_{12} \hat{e}_1 + \Omega_{23} \hat{e}_3 = -\omega_3 \hat{e}_1 + \omega_1 \hat{e}_3 = \vec{\omega} \times \hat{e}_2 \quad \checkmark$$

$$\vec{\omega} \times \hat{e}_2 = (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2 + \omega_3 \hat{e}_3) \times \hat{e}_2 = \omega_1 \hat{e}_3 - \omega_3 \hat{e}_1$$

$$\hat{e}_1 \times \hat{e}_2 = \hat{e}_3 \quad \hat{e}_3 \times \hat{e}_2 = -\hat{e}_1$$

$$\frac{d\hat{e}_3}{dt} = \vec{\omega} \times \hat{e}_3$$

## Rotating reference frames



Fixed inertial reference frame  $\hat{e}_1^0, \hat{e}_2^0, \hat{e}_3^0$

Rotating (non-inertial) reference frame  $\hat{e}_1, \hat{e}_2, \hat{e}_3$

$$\left. \frac{d\vec{b}}{dt} \right|_{\text{inertial}} = \frac{d}{dt} \left( \sum_{i=1}^3 b_i^0 \hat{e}_i^0 \right) = \sum_{i=1}^3 \dot{b}_i^0 \hat{e}_i^0$$

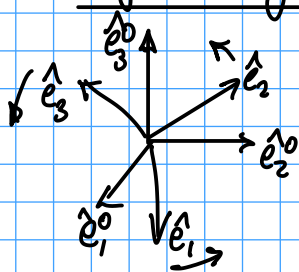
$$\left. \frac{d\vec{b}}{dt} \right|_{\text{inertial}} = \frac{d}{dt} \left( \sum_{i=1}^3 b_i \hat{e}_i \right) = \sum_{i=1}^3 \dot{b}_i \hat{e}_i + \sum_{i=1}^3 b_i \dot{\hat{e}}_i \quad \left. \frac{d\vec{b}}{dt} \right|_{\text{body}} = \sum_{i=1}^3 \frac{db_i}{dt} \hat{e}_i$$

$$\dot{\hat{e}}_i = \frac{d\hat{e}_i}{dt} = \vec{\omega} \times \hat{e}_i$$

For rotations  $\left. \frac{d\vec{b}}{dt} \right|_{\text{inertial}} = \left. \frac{d\vec{b}}{dt} \right|_{\text{body}} + \sum_{i=1}^3 b_i (\vec{\omega} \times \hat{e}_i)$

$$\boxed{\left. \frac{d\vec{b}}{dt} \right|_{\text{inertial}} = \left. \frac{d\vec{b}}{dt} \right|_{\text{body}} + \vec{\omega} \times \vec{b}}$$

## Rigid body rotations



$\vec{r}_p$ : position of point  $p$  in the body frame (CONSTANT IN BODY FRAME)

Rigid body with one fixed point.

$$T = \sum_p \frac{1}{2} m_p v_p^2 = \sum_p \frac{1}{2} m_p (\vec{\omega} \times \vec{r}_p)^2$$

Levi-Civita symbol  
 $\epsilon_{ijk} = \begin{cases} +1 & \text{ijk cyc 123} \\ -1 & \text{ijk cyc 213} \\ 0 & \text{if any repeated index} \end{cases}$

$$\epsilon_{123} = \epsilon_{231} = \epsilon_{312} = +1$$

$$\epsilon_{213} = \epsilon_{132} = \epsilon_{321} = -1$$

$$\epsilon_{jji} = \epsilon_{ijj} = \epsilon_{jij} = 0$$

$$\hat{e}_1 \times \hat{e}_2 = \hat{e}_3 = \sum_l \epsilon_{12l} \hat{e}_l \checkmark$$

$$\sum_l \epsilon_{12l} \hat{e}_l = \hat{e}_3$$

$$\hat{e}_i \times \hat{e}_j = \sum_l \epsilon_{ijl} \hat{e}_l \left( \underset{\text{NOTATION}}{=} \epsilon_{ijl} \hat{e}_l \right)$$

$$\vec{a} \times \vec{b} = \left( \sum_i a_i \hat{e}_i \right) \times \left( \sum_j b_j \hat{e}_j \right) = \sum_{ijl} a_i b_j \epsilon_{ijl} \hat{e}_l = \sum_{ijk} \epsilon_{ijk} a_i b_j \hat{e}_k \quad (\vec{a} \times \vec{b})_k = \sum_{ij} \epsilon_{ijk} a_i b_j$$

Also  $\sum_i \epsilon_{jki} \epsilon_{ilm} = \sum_i \epsilon_{ijk} \epsilon_{ilm} = \delta_{jm} \delta_{kl} - \delta_{jm} \delta_{ek}$

$\epsilon_{ijk} = \epsilon_{jki} = \epsilon_{kij}$  (CYCLIC)

$$\begin{aligned} (\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) &= \left( \sum_{ijk} A_i B_j \epsilon_{ijk} \hat{e}_k \right) \cdot \left( \sum_{lmn} C_l D_m \epsilon_{lmn} \hat{e}_n \right) = \sum_{ijklm} \epsilon_{ijk} \epsilon_{lmn} A_i B_j C_l D_m \\ &= \sum_{ijklm} (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) A_i B_j C_l D_m = (\vec{A} \cdot \vec{C})(\vec{B} \cdot \vec{D}) - (\vec{A} \cdot \vec{D})(\vec{B} \cdot \vec{C}) \end{aligned}$$

## IN SUMMARY:

$$T = \sum_P \frac{1}{2} m_P v_P^2 = \sum_P \frac{1}{2} m_P (\vec{\omega} \times \vec{r}_P)^2 = \sum_P \frac{m_P}{2} \left[ \omega^2 r_P^2 - (\vec{\omega} \cdot \vec{r}_P)^2 \right]$$

$$(\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) = (\vec{A} \cdot \vec{C})(\vec{B} \cdot \vec{D}) - (\vec{A} \cdot \vec{D})(\vec{B} \cdot \vec{C}) \Rightarrow (\vec{\omega} \times \vec{r}_P) \cdot (\vec{\omega} \times \vec{r}_P) = (\vec{\omega} \cdot \vec{\omega})(\vec{r}_P \cdot \vec{r}_P) - (\vec{\omega} \cdot \vec{r}_P)(\vec{\omega} \cdot \vec{r}_P)$$

$$T = \frac{1}{2} \sum_P m_P \left[ r_P^2 \sum_{i=1}^3 \omega_i \omega_i - \left( \sum_{i=1}^3 \omega_i r_{Pi} \right) \left( \sum_{j=1}^3 \omega_j r_{Pj} \right) \right] = \frac{1}{2} \sum_P m_P \sum_{ij=1}^3 \left[ r_P^2 \delta_{ij} - r_{Pi} r_{Pj} \right] \omega_i \omega_j$$

$$T = \frac{1}{2} \sum_{ij=1}^3 I_{ij} \omega_i \omega_j$$

$$I_{ij} = \sum_P m_P \left[ r_P^2 \delta_{ij} - r_{Pi} r_{Pj} \right]$$

Moment of inertia tensor (SYMMETRIC)

$$\vec{L} = \sum_P m_P \vec{r}_P \times \vec{v}_P = \sum_P m_P \vec{r}_P \times (\vec{\omega} \times \vec{r}_P)$$

Aside: Using summation notation:

$$\begin{aligned} [A \times (B \times C)]_i &= \epsilon_{ijk} A_j \epsilon_{k\ell m} B_\ell C_m = \epsilon_{kij} \epsilon_{k\ell m} A_j B_\ell C_m = (\delta_{i\ell} \delta_{jm} - \delta_{im} \delta_{j\ell}) A_j B_\ell C_m \\ &= (\vec{A} \cdot \vec{C}) B_i - (\vec{A} \cdot \vec{B}) C_i \end{aligned}$$

$$[\vec{r}_P \times (\vec{\omega} \times \vec{r}_P)]_i = (\vec{r}_P \cdot \vec{r}_P) \omega_i - (\vec{r}_P \cdot \vec{\omega}) r_{Pi}$$

$$L_i = \sum_P m_P (\vec{r}_P \times \vec{v}_P)_i = \sum_P m_P [\vec{r}_P \times (\vec{\omega} \times \vec{r}_P)]_i = \sum_P m_P \left[ r_P^2 \omega_i - \sum_{j=1}^3 r_{Pi} r_{Pj} \omega_j \right]$$

$$L_i = \sum_{j=1}^3 \left( \sum_P m_P [r_P^2 \delta_{ij} - r_{Pi} r_{Pj}] \right) \omega_j = \sum_{j=1}^3 I_{ij} \omega_j$$

$$L_i = \sum_j I_{ij} \omega_j$$

$$T = \frac{1}{2} \vec{\omega} \cdot \vec{L}$$

$$T = \frac{1}{2} \sum_{ij=1}^3 I_{ij} \omega_i \omega_j$$

$$I_{ij} = I_{ji} = I_{ji}^* \Rightarrow \text{HERMITIAN} \Rightarrow \text{CAN BE DIAGONALIZED} \quad \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}$$

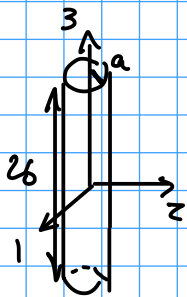
Directions that diagonalize  $I_{ij}$  are called **PRINCIPAL AXES**

### EXAMPLES

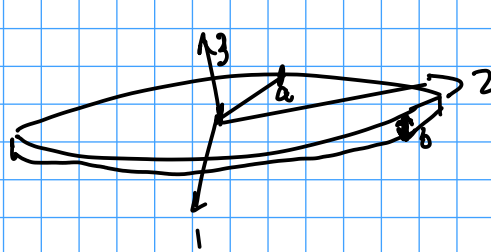
Homogeneous, CUBE OR SPHERE



$$\begin{pmatrix} I & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{pmatrix} = I \mathbf{1}_{3 \times 3}$$



$$a \ll b \Rightarrow I_1 = I_2 \gg I_3$$



$$a \gg b \Rightarrow I_1 = I_2 < I_3$$

$$I = \sum_p m_p (\text{distance to axis})^2 = \sum_p m_p r_{p\perp}^2$$

$$I_3 \sim M a^2 \quad I_1 \sim M b^2$$

$$a \ll b \Rightarrow I_3 \ll I_1$$

$$a \gg b \Rightarrow I_3 \gg I_1$$

More irregular:  $I_1, I_2, I_3$  all different

$$\begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix} \Leftrightarrow I_{ij} = I_i \delta_{ij} \quad L_i = \sum_{j=1}^3 I_{ij} \omega_j = \sum_{j=1}^3 I_i \delta_{ij} \omega_j = I_i \omega_i$$

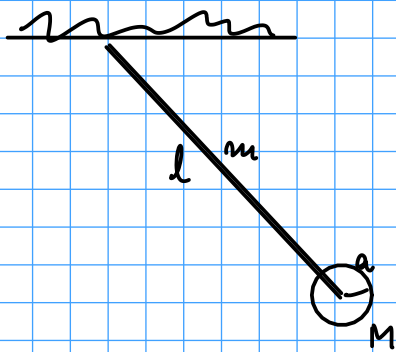
$$\begin{pmatrix} L_1 \\ L_2 \\ L_3 \end{pmatrix} = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix} = \begin{pmatrix} I_1 \omega_1 \\ I_2 \omega_2 \\ I_3 \omega_3 \end{pmatrix}$$

$$T = \frac{1}{2} \vec{\omega} \cdot \vec{L} = \frac{1}{2} \sum_{i=1}^3 \omega_i L_i = \frac{1}{2} \sum_{i=1}^3 \omega_i I_i \omega_i = \frac{1}{2} \sum_{i=1}^3 I_i \omega_i^2$$

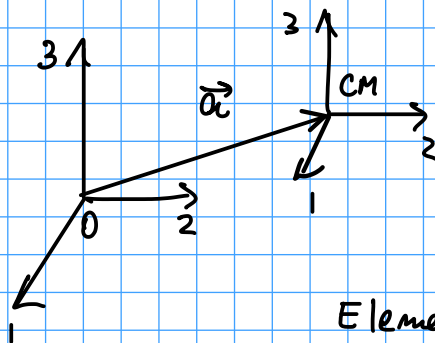
$$L_i = I_i \omega_i$$

$$T = \frac{1}{2} \sum_{i=1}^3 I_i \omega_i^2$$

# Parallel Axis Theorem



( $\bar{I}_{ij}$  MoI of I wrt CM)



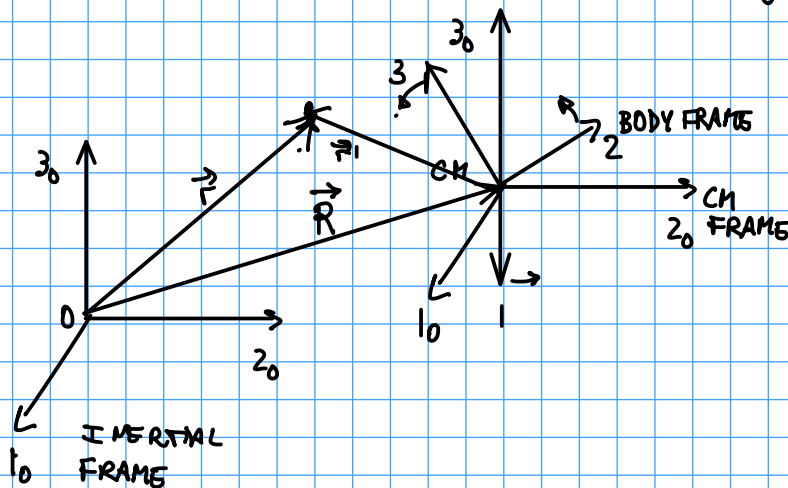
$$I_{ij} = \bar{I}_{ij} + M[\delta_{ij}a^2 - a_i a_j]$$

$\updownarrow$  ORIGIN O       $\updownarrow$  ORIGIN CM

Elementary version:  $I = \bar{I} + M d^2$

## Rotation wrt an arbitrarily moving point

No longer look at a point in the body fixed in space



• INERTIAL FRAME

• CM FRAME { MAY NOT BE INERTIAL IF  $\ddot{\vec{R}} \neq 0$   
 AXES // AXES OF INERTIAL FRAME

• BODY FRAME { ORIGIN AT  $\vec{R} \Rightarrow$  NON-INERTIAL  $\ddot{\vec{R}} \neq 0$   
 AXES 123 ROTATING WRT  $1_0 2_0 3_0$

$$\vec{r} = \vec{R} + \vec{r}'$$

$$\left. \frac{d\vec{r}}{dt} \right|_{\text{INERTIAL}} = \frac{d\vec{R}}{dt} + \left. \frac{d\vec{r}'}{dt} \right|_{\text{CM}}$$

$$\left. \frac{d\vec{r}}{dt} \right|_{\text{INERTIAL}} = \frac{d\vec{R}}{dt} + \left. \frac{d\vec{r}'}{dt} \right|_{\text{BODY}} + \vec{\omega} \times \vec{r}'$$

IF RIGID BODY  $\left. \frac{d\vec{r}}{dt} \right|_{\text{INERTIAL}} = \frac{d\vec{R}}{dt} + \vec{\omega} \times \vec{r}'$

For any system of particles:

$$M = \sum_p m_p$$

$$\vec{R} \equiv \text{CENTER OF MASS POSITION} = \frac{\sum_p m_p \vec{r}_p}{\sum_p m_p}$$

$$T = \frac{1}{2} M \dot{\vec{R}}^2 + T'$$

$$T' = \sum_p \frac{m_p}{2} \dot{\vec{r}}_p'^2$$

MEASURED IN FRAME WITH ORIGIN AT CM (CM FRAME)

$$\vec{L} = \vec{R} \times M \dot{\vec{R}} + \vec{L}'$$

$$\vec{L}' = \sum_p \vec{r}_p' \times m_p \dot{\vec{r}}_p'$$

For rigid body  $\dot{\vec{r}}_p' = \vec{\omega} \times \vec{r}_p'$

$$\vec{L} = \vec{R} \times M \dot{\vec{R}} + \underbrace{\sum_p \vec{r}_p' \times m_p (\vec{\omega} \times \vec{r}_p')}_{\text{CALCULATED BEFORE}}$$

$$L_i = \sum_j \bar{I}_{ij} \omega_j$$

$$T = \frac{1}{2} M |\dot{\vec{R}}|^2 + \sum_p \frac{m_p}{2} (\vec{\omega} \times \vec{r}_p')^2$$

CALCULATED BEFORE

$$T' = \frac{1}{2} \sum_{ij} \bar{I}_{ij} \omega_i \omega_j$$

# EULER'S EQUATIONS

$$\frac{d\vec{L}}{dt} = \vec{N} \quad \text{IF} \left\{ \begin{array}{l} \times \text{ INERTIAL FRAME} \\ \times \text{ CM FRAME (ORIGIN AT CM)} \\ \text{MAY BE ACCELERATED BUT} \\ \text{IT STAYS PARALLEL TO INERTIAL FRAME} \end{array} \right\} \Leftrightarrow \text{"TRANSLATED BUT NOT ROTATED"}$$

$$\vec{N} = \frac{d\vec{L}}{dt} = \left. \frac{d\vec{L}}{dt} \right|_{\text{BODY}} + \vec{\omega} \times \vec{L}$$

$$N_1 = \left. \frac{dL_1}{dt} \right|_{\text{BODY}} + (\vec{\omega} \times \vec{L})_1 = \left. \frac{dL_1}{dt} \right|_{\text{BODY}} + \omega_2 L_3 - \omega_3 L_2$$

$$L_i = I_i \omega_i \quad \dot{I}_i|_{\text{BODY}} = 0 \quad (\text{RIGID BODY} \Rightarrow \vec{r}_P = \vec{r} \text{ IN BODY FRAME} \Rightarrow \dot{I}_i|_{\text{BODY}} = 0)$$

$$= I_1 \left. \frac{d\omega_1}{dt} \right|_{\text{BODY}} + \omega_2 I_3 \omega_3 - \omega_3 I_2 \omega_2 = I_1 \left. \frac{d\omega_1}{dt} \right|_{\text{BODY}} + \omega_2 \omega_3 (I_3 - I_2)$$

$$I_1 \left. \frac{d\omega_1}{dt} \right|_{\text{BODY}} = \omega_2 \omega_3 (I_2 - I_3) + N_1$$

$$I_2 \left. \frac{d\omega_2}{dt} \right|_{\text{BODY}} = \omega_3 \omega_1 (I_3 - I_1) + N_2$$

$$I_3 \left. \frac{d\omega_3}{dt} \right|_{\text{BODY}} = \omega_1 \omega_2 (I_1 - I_2) + N_3$$

EULER'S  
EQUATIONS

## APPLICATIONS

Kater's Pendulum

$n=1$  degree of freedom  $\Leftrightarrow$  angle  $\Phi \equiv$  gen coord

NO FRICTION

In Lab frame:

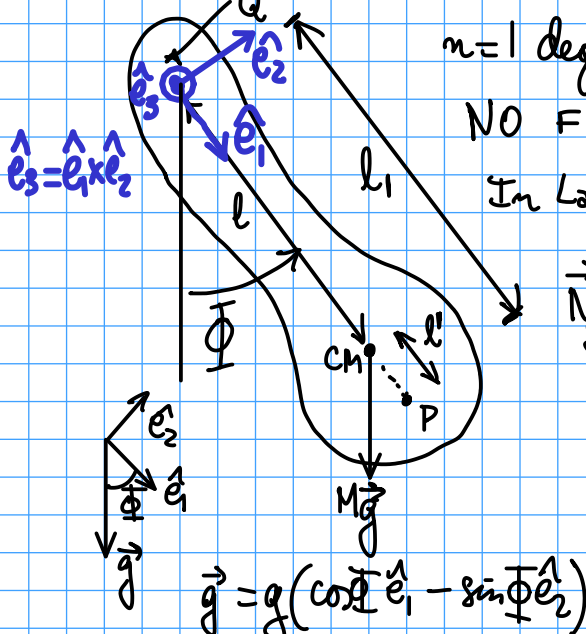
$$I \dot{\omega}_3 = \frac{dL_3}{dt} = N_3 = -Mg l \sin \Phi \quad \omega_3 = \dot{\Phi}$$

$$\vec{N}_g = l \hat{e}_1 \times M \vec{g} = l \hat{e}_1 \times Mg (\cos \Phi \hat{e}_1 - \sin \Phi \hat{e}_2) = -Mg l \sin \Phi \hat{e}_3$$

SMALL ANGLE  $\sin \Phi \approx \Phi$

$$I \ddot{\Phi} \approx -Mg l \Phi \quad \ddot{\Phi} \approx -\frac{Mg l}{I} \Phi$$

$$\ddot{\Phi} = -\Omega^2 \Phi \quad \Omega = \left( \frac{Mg l}{I} \right)^{1/2} \text{ FREQ. OF OSCILLATION}$$



## Parallel axis theorem

$$I = \bar{I} + M l^2 = M \bar{k}^2 + M l^2 = M (\bar{k}^2 + l^2) = M k^2$$

DEFINE RADIUS OF GYRATION  $k$

BY  $I = M k^2$

DEFINE  $l_1 = \frac{k^2}{l} = l + \frac{\bar{k}^2}{l}$

$$\Omega_Q^2 = \frac{M g l}{M (\bar{k}^2 + l^2)} = \frac{g}{l_1}$$

$$[\Omega^2] = \frac{\cancel{M} / \cancel{M}}{\cancel{L} / L^2} = \frac{1}{L^2} \Rightarrow [\Omega] = \frac{1}{L}$$

COMPARE WITH SIMPLE PENDULUM  $\Omega = \sqrt{\frac{g}{L}}$

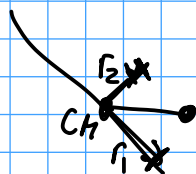


point mass

$$\bar{I} = 0 \Rightarrow \bar{k} = 0 \quad l_1 = l + \frac{\bar{k}^2}{l} = l$$

$$\bar{I} = \int d^3 r \rho(\vec{r}) (r_1^2 + r_2^2)$$

distances measured wrt CM



POINT MASS

$$r_1 = r_2 = 0$$

$$\bar{k}^2 = \frac{\bar{I}}{M} = \frac{\int d^3 r \rho(\vec{r}) (r_1^2 + r_2^2)}{\int d^3 r \rho(\vec{r})}$$

↔ Average squared distance to axis of rotation weighted by the mass density  $\rho(\vec{r})$

Suspend the pendulum from point P "Percussion center" at distance  $l_1$  from Q, along the direction that goes through CM

$$\Omega_Q^2 = \frac{M g l}{M (\bar{k}^2 + l^2)} = \frac{g}{l_1}$$

$$\Omega_P^2 = \frac{g l_1}{l_1^2} = \frac{g (l_1 - l)}{l_1^2 + (l_1 - l)^2} = \frac{g \bar{k}^2 / l}{\bar{k}^2 + \bar{k}^4 / l^2} = \frac{g l}{l^2 + \bar{k}^2} = \Omega_Q^2$$

$$l_1 = l + \frac{\bar{k}^2}{l} \quad l_1 - l = \frac{\bar{k}^2}{l} \quad (l_1 - l)^2 = \frac{\bar{k}^4}{l^2}$$

## Torque-Free Motion in 3D: SYMMETRIC TOP

$$I_1 = I_2 \neq I_3 \quad \Omega \equiv \omega_3 \frac{I_3 - I_1}{I_1}$$

$$I_1 \left. \frac{d\omega_1}{dt} \right|_{\text{BODY}} = \omega_2 \omega_3 (I_2 - I_3) \Rightarrow \dot{\omega}_1 = - \left[ \omega_3 \frac{(I_3 - I_2)}{I_1} \right] \omega_2 \Rightarrow$$

$$\dot{\omega}_1 = -\Omega \omega_2$$

$$I_2 \left. \frac{d\omega_2}{dt} \right|_{\text{BODY}} = \omega_3 \omega_1 (I_3 - I_1) \Rightarrow \dot{\omega}_2 = \left[ \omega_3 \frac{(I_3 - I_1)}{I_2} \right] \omega_1 \Rightarrow$$

$$\dot{\omega}_2 = \Omega \omega_1$$

$$I_3 \left. \frac{d\omega_3}{dt} \right|_{\text{BODY}} = \omega_1 \omega_2 (I_1 - I_2) \Rightarrow$$

$$\dot{\omega}_3 = 0$$

$$\dot{\vec{\omega}} = \Omega \hat{e}_3 \times \vec{\omega} \Leftrightarrow \vec{\omega} \text{ rotates with angular velocity } \Omega \hat{e}_3 \Rightarrow \text{PRECESSION}$$

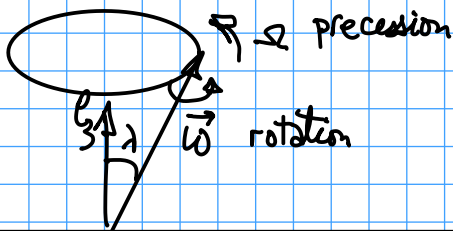
$$\dot{\vec{\omega}} = \Omega \hat{e}_3 \times \vec{\omega} = \Omega \hat{e}_3 \times (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2 + \omega_3 \hat{e}_3) = \Omega (\omega_1 \hat{e}_2 - \omega_2 \hat{e}_1)$$

$$\dot{\omega}_1 = -\Omega \omega_2 \quad \dot{\omega}_2 = \Omega \omega_1 \quad \dot{\omega}_3 = 0$$

$\vec{\omega}$  rotates around  $\hat{e}_3$  with ang speed  $\Omega$

If initially  $\vec{\omega} = \omega (\sin \lambda \hat{e}_1 + \cos \lambda \hat{e}_3)$   $\vec{\omega} \in \text{plane of } \hat{e}_1 \text{ and } \hat{e}_3$

$$\omega_1 = \omega \sin \lambda \cos(\Omega t) \quad \omega_2 = \omega \sin \lambda \sin(\Omega t) \quad \omega_3 = \omega \cos \lambda$$



$\hat{e}_1^0, \hat{e}_2^0, \hat{e}_3^0$  INERTIAL SET OF UNIT VECTORS  
(NOT ROTATING)

$\hat{e}_1, \hat{e}_2, \hat{e}_3$  BODY SET OF UNIT VECTORS  
(ROTATING WITH BODY)

We need 6 coordinates to describe positions of all points  
in a RIGID BODY

CM POSITION  $\vec{R} \leftrightarrow X Y Z$

3 coord to fix orientation of  
 $\hat{e}_1, \hat{e}_2, \hat{e}_3$  wrt  $\hat{e}_1^0, \hat{e}_2^0, \hat{e}_3^0$

WE CHOOSE EULER ANGLES:  $\alpha, \beta, \gamma$

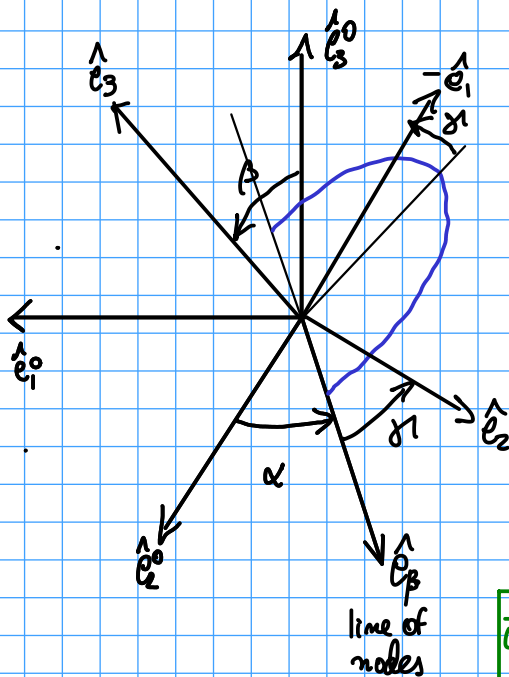
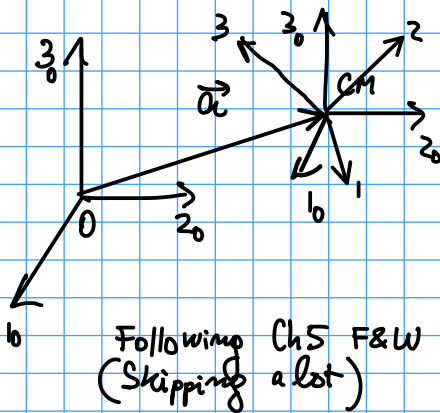
- $\hat{e}_\beta \perp \hat{e}_\alpha^0, \hat{e}_\gamma^0$
- ORIENTATION OF NEW NORTH POLE  $\rightarrow \beta$  POLAR  $\rightarrow \alpha$  AZIMUTHAL
- $\gamma$  SPECIFIES ADDITIONAL ROTATION AROUND  $\hat{e}_\beta$
- $\alpha, \beta, \gamma$  CAN BE CHOSEN INDEPENDENTLY AND THEY COMPLETELY SPECIFY ORIENTATION OF BODY FRAME  $\Rightarrow$  USE  $\alpha, \beta, \gamma$  AS GENERALIZED COORD

$$T = \frac{1}{2} (\mathcal{I}_1 \omega_1^2 + \mathcal{I}_2 \omega_2^2 + \mathcal{I}_3 \omega_3^2) \quad (1)$$

$$\vec{\omega} = \alpha \hat{e}_3^0 + \beta \hat{e}_\beta + \gamma \hat{e}_\gamma \quad (2) \quad \hat{e}_\beta = \cos \gamma \hat{e}_2 + \sin \gamma \hat{e}_1 \quad (3)$$

$$\hat{e}_\gamma^0 = \cos \beta \hat{e}_3 + \sin \beta [\cos \gamma (-\hat{e}_1) + \sin \gamma \hat{e}_2] \quad (4)$$

$$\vec{\omega} = (-\alpha \sin \beta \cos \gamma + \beta \sin \gamma) \hat{e}_1 + (\alpha \sin \beta \sin \gamma + \beta \cos \gamma) \hat{e}_2 + (\gamma + \alpha \cos \beta) \hat{e}_3 \quad (5)$$



NOTE: Infinitesimal rot  $R_k = 1 + M_k \dot{\alpha} dt$  (6)  $R_\alpha R_\beta = 1 + (M_\alpha \dot{\alpha} + M_\beta \dot{\beta}) dt + \frac{1}{2} \frac{d^2}{dt^2}$  (8) EQUAL TO ORDER  $dt$   
 $R_\beta = 1 + M_\beta \dot{\beta} dt$  (7)  $R_\beta R_\alpha = 1 + (M_\beta \dot{\beta} + M_\alpha \dot{\alpha}) dt + \frac{1}{2} \frac{d^2}{dt^2}$  (9)  
 $\Rightarrow$  INFINITESIMAL ROTATIONS ADD,  $\vec{\omega} = \vec{\omega}_\alpha + \vec{\omega}_\beta + \vec{\omega}_\gamma$  (10)  $\Rightarrow$  (2)

## KINETIC ENERGY

General case:

$$T = \frac{I_1}{2} \omega_1^2 + \frac{I_2}{2} \omega_2^2 + \frac{I_3}{2} \omega_3^2 = \frac{I_1}{2} (-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma)^2 + \frac{I_2}{2} (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma)^2 + \frac{I_3}{2} (\dot{\gamma} + \dot{\alpha} \cos \beta)^2 \quad (11)$$

Symmetric Top:  $I_1 = I_2$  (12)  $T = \frac{I_1}{2} (\dot{\beta}^2 + \dot{\alpha}^2 \sin^2 \beta) + \frac{I_3}{2} (\dot{\gamma} + \dot{\alpha} \cos \beta)^2$  (13)

Torque-Free Motion of symmetric top  $L = T = \frac{I_1}{2} (\dot{\beta}^2 + \dot{\alpha}^2 \sin^2 \beta) + \frac{I_3}{2} (\dot{\gamma} + \dot{\alpha} \cos \beta)^2$  (14)

$\alpha, \beta, \gamma$  Gen Coord  $\Rightarrow$  3 Lagrange Eqs

2 CYCLIC COORDINATES:  $\alpha, \gamma$  (15)

WE WILL SHOW LATER THAT:

$$p_\alpha = \vec{L} \cdot \hat{e}_3^0 \quad (18)$$

$$p_\gamma = \vec{L} \cdot \hat{e}_3 \quad (19)$$

$\Rightarrow$  2 CONST OF MOTION  $\left\{ \begin{array}{l} p_\alpha = \frac{\partial L}{\partial \dot{\alpha}} = I_1 \sin^2 \beta \dot{\alpha} + I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta \quad (16) \\ p_\gamma = \frac{\partial L}{\partial \dot{\gamma}} = I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \quad (17) \end{array} \right.$

Recall  $\vec{\omega} = (-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma) \hat{e}_1 + (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma) \hat{e}_2 + (\dot{\gamma} + \dot{\alpha} \cos \beta) \hat{e}_3$  (5)

We compute now  $p_\gamma = \frac{\partial L}{\partial \dot{\gamma}} = I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta)$  (17) and notice that  $p_\gamma \stackrel{(5)}{=} I_3 \omega_3 = \vec{L} \cdot \hat{e}_3$  (19) ✓ AS ANTICIPATED

Also  $p_\alpha = \frac{\partial L}{\partial \dot{\alpha}} = I_1 \sin^2 \beta \dot{\alpha} + I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta$  (16)  $\vec{L} = I_1 (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2) + I_3 \omega_3 \hat{e}_3$  (20)

Compare with  $\vec{L} \cdot \hat{e}_3^0 = (I_1 (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2) + I_3 \omega_3 \hat{e}_3) (\cos \beta \hat{e}_3 + \sin \beta [\cos \gamma (-\hat{e}_1) + \sin \gamma \hat{e}_2])$  (21)

$$= I_1 [(-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma) (-\sin \beta \cos \gamma) + (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma) (\sin \beta \sin \gamma)] + I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta \quad (22)$$

$$= I_1 \dot{\alpha} \sin^2 \beta (\cos^2 \gamma + \sin^2 \gamma) + I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta \quad (23)$$

$$\boxed{\vec{L} \cdot \hat{e}_3^0 = I_1 \dot{\alpha} \sin^2 \beta + I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta = p_\alpha} \quad \checkmark \quad (18)$$

$p_\beta = \frac{\partial L}{\partial \dot{\beta}} = I_1 \dot{\beta}$  (24) NOT A CONSTANT (IN PRINCIPLE) BECAUSE  $\beta$  APPEARS EXPLICITLY IN THE LAGRANGIAN

$$\frac{\partial L}{\partial \dot{\beta}} = I_1 \sin \beta \cos \beta \dot{\alpha}^2 - I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \dot{\alpha} \sin \beta = \dot{\alpha} \sin \beta (I_1 \dot{\alpha} \cos \beta - I_3 \omega_3) \quad (25)$$

Then 
$$\dot{p}_\beta = \frac{\partial L}{\partial \beta} = \dot{\alpha} \sin \beta (I_1 \dot{\alpha} \cos \beta - I_3 \omega_3) \quad (26) \quad (\text{However, see later})$$

Compare  $p_\beta$  with 
$$\vec{L} \cdot \hat{e}_\beta = \left( I_1 (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2) + I_3 \omega_3 \hat{e}_3 \right) (\cos \gamma \hat{e}_2 + \sin \gamma \hat{e}_1)$$
 (20) (3)

Recall 
$$\vec{\omega} = (-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma) \hat{e}_1 + (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma) \hat{e}_2 + (\dot{\gamma} + \dot{\alpha} \cos \beta) \hat{e}_3 \quad (5)$$

$$\begin{aligned} \vec{L} \cdot \hat{e}_\beta &= I_1 [(-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma) \sin \gamma + (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma) \cos \gamma] \quad (27) \\ &= I_1 \dot{\beta} (\sin^2 \gamma + \cos^2 \gamma) = I_1 \dot{\beta} \Rightarrow \boxed{\vec{L} \cdot \hat{e}_\beta = p_\beta} \quad (28) \quad (29) \quad \checkmark \end{aligned}$$

### Description in Inertial (or CM) Frame

No torque  $\Rightarrow \dot{\vec{L}} = 0$  (30)  $\Rightarrow$  CHOOSE  $\hat{e}_3^0 // \vec{L}$  (32)  $\Rightarrow \vec{L} \cdot \hat{e}_3^0 = |\vec{L}|$  AND  $\vec{L} = |\vec{L}| \hat{e}_3^0$  (35)

(31)  $\Rightarrow |\vec{L}|$  doesn't change (33)

We know that  $\vec{L} \cdot \hat{e}_3 = p_\beta$  is a constant of the motion

$$\underbrace{\vec{L} \cdot \hat{e}_3}_{= p_\beta} = \underbrace{|\vec{L}|}_{= \text{CONST}} \underbrace{\hat{e}_3^0 \cdot \hat{e}_3}_{= \text{CONST}} \Rightarrow \hat{e}_3^0 \cdot \hat{e}_3 \text{ CONSTANT} \quad (36) \quad (37)$$

CONST =  $\hat{e}_3^0 \cdot \hat{e}_3 = \cos(\text{angle between } \hat{e}_3 \text{ and } \hat{e}_3^0) = \cos \beta$  (38)

$\beta$  IS A CONSTANT OF THE MOTION  $\Rightarrow \dot{\beta} = 0$  (40)

For this particular choice of axes,  $p_\beta$  and  $\vec{L} \cdot \hat{e}_\beta$  are zero!!

$$p_\beta = I_1 \dot{\beta} = \vec{L} \cdot \hat{e}_\beta = 0 \quad (41)$$

(41)  $\vec{L} \cdot \hat{e}_\beta = 0$   
 $\vec{L} \neq 0 \quad |\hat{e}_\beta| = 1 \neq 0 \Rightarrow \vec{L} \perp \hat{e}_\beta \Rightarrow \vec{L}$  is in the plane of  $\hat{e}_2$  and  $\hat{e}_3$  (42) (43)

CONSTANTS:  $I_1, I_3, \beta, I_3 \omega_3 = L_3, p_\beta$

$$p_\alpha = \frac{\partial L}{\partial \dot{\alpha}} = I_1 \sin^2 \beta \dot{\alpha} + I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta \stackrel{(16)}{=} \stackrel{(5)}{=} \underbrace{I_1 \sin^2 \beta}_{(47)} \dot{\alpha} + \underbrace{I_3 \omega_3 \cos \beta}_{(44)} \Rightarrow \dot{\alpha} \text{ CONSTANT} \quad (45)$$

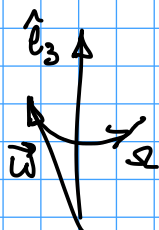
$$p_\gamma = \frac{\partial L}{\partial \dot{\gamma}} = I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \quad (17)$$

$$\left. \begin{array}{l} I_3, p_\beta, \dot{\alpha}, \beta \text{ CONSTANT} \end{array} \right\} \Rightarrow \dot{\gamma} \text{ CONSTANT} \quad (46)$$

$\alpha = \dot{\alpha} t \quad \gamma = \dot{\gamma}_0 t$   
 (47)  $\uparrow$  CONSTANTS  $\uparrow$  (48)

$$(4) \Rightarrow \dot{p}_\beta = \frac{\partial L}{\partial \beta} \stackrel{(26)}{=} \dot{\alpha} \sin \beta (I_1 \dot{\alpha} \cos \beta - I_3 \omega_3) \Rightarrow \boxed{\dot{\alpha} \cos \beta = \frac{I_3 \omega_3}{I_1}} \quad (49)$$

$$(5) \Rightarrow \omega_3 = \dot{\gamma} + \dot{\alpha} \cos \beta \stackrel{(49)}{=} \dot{\gamma} + \frac{I_3}{I_1} \omega_3 \Rightarrow \boxed{\dot{\gamma} = \omega_3 \frac{I_1 - I_3}{I_1} = -\Omega} \quad (51)$$



$\Omega \equiv \omega_3 \frac{I_3 - I_1}{I_1} \stackrel{(51)}{\Rightarrow}$  defined when looking at body frame evolution of  $\vec{\omega}$  a couple of lectures before

$$(2) \Rightarrow \vec{\omega} = \dot{\alpha} \hat{e}_3^0 + \dot{\gamma} \hat{e}_3 \stackrel{(52)}{\Rightarrow} \boxed{\hat{e}_3, \hat{e}_3^0, \vec{\omega}, \vec{L} \text{ are in one plane}} \quad (53)$$

Consider cases where  $I_1 = I_2 < I_3$  or  $I_1 = I_2 > I_3$

$$\vec{L} = I_1 (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2) + I_3 \omega_3 \hat{e}_3 \quad (20)$$

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SIMPLE PENDULUM:

$$\ddot{\theta} = -\frac{g}{l} \sin \theta$$

THUMBTRACK:

$$(\text{angle})'' = - \left[ \begin{matrix} \text{AWFUL} \\ \text{MESS} \end{matrix} \right] \sin(\text{angle})$$


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